

operating points are NOT fixed:

batch · context · KV precision · kernel/hardware

all move a point across the ridge

**memory-bound
(bandwidth)**

**compute-bound
(FLOPs)**

decode @ low batch

prefill @ 9.2K prompt

**roofline
ridge ~295**

**HBM weight-stream per token:
~1 FLOP/byte at batch-1 (Ch8)**

**2·N·L over HBM weight stream:
~9.2K FLOP/byte (high intensity)**

lower →

→ higher

arithmetic intensity (FLOP/byte) →